



D4 Genetics and inheritance

D · Continuity and change · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Alleles are alternative versions of a gene at the same locus; an organism's genotype (its combination of alleles) interacts with the environment to produce its phenotype (observable characteristics). Mendelian inheritance patterns include simple dominant/recessive inheritance, codominance, multiple alleles, and sex-linkage (genes located on sex chromosomes, typically X); crosses are predicted using Punnett squares, based on the independent assortment of alleles, giving characteristic ratios such as 3:1 for a monohybrid cross between two heterozygotes, or 9:3:3:1 for a dihybrid cross. Pedigree charts allow inheritance patterns, including sex-linked and autosomal recessive conditions, to be traced through families.

Worked example: follow the reasoning

Two heterozygous pea plants ($Tt \times Tt$) for stem height, where T (tall) is dominant to t (dwarf), are crossed. Step 1: each parent can produce gametes carrying either T or t, in equal proportions. Step 2: construct a Punnett square: possible offspring genotypes are TT, Tt, Tt, tt, in equal 1:2:1 proportions. Step 3: since T is dominant, both TT and Tt genotypes give a tall phenotype, while only tt gives a dwarf phenotype, so the phenotype ratio is 3 tall : 1 dwarf. Step 4: the probability that a given offspring is heterozygous (a 'carrier', Tt) is 2 out of 4, i.e. 50%. Conclusion: this cross gives an expected 3:1 tall-to-dwarf phenotype ratio, with 50% of offspring being heterozygous carriers.

Quick check · multiple choice

In a dihybrid cross between two individuals heterozygous for two independently assorting genes ($AaBb \times AaBb$), what phenotypic ratio is expected among the offspring?

- A 3:1
- B 1:2:1
- C 9:3:3:1
- D 1:1:1:1

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A particular form of colour blindness is caused by a recessive allele (b) located on the X chromosome; the dominant allele (B) gives normal colour vision. A colour-blind man has children with a woman who is a heterozygous carrier for this condition. [8]
 - (a) State the genotypes of the man and the woman using the alleles given. [2]
 - (b) Construct a Punnett square (or describe the cross) and determine the expected proportion of daughters and sons who will be colour-blind. [4]
 - (c) Explain why colour blindness caused by this X-linked recessive allele is more common in males than in females in the general population. [2]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) The colour-blind man is $X^b Y$; the carrier woman is $X^B X^b$.

(b) Cross $X^b Y \times X^B X^b$: possible offspring are $X^B X^b$ (carrier daughter, normal vision), $X^b X^b$ (colour-blind daughter), $X^B Y$ (son, normal vision), $X^b Y$ (colour-blind son), each with probability $1/4$. So $1/2$ of daughters are expected to be colour-blind ($X^b X^b$) and $1/2$ unaffected carriers; $1/2$ of sons are expected to be colour-blind ($X^b Y$) and $1/2$ unaffected ($X^B Y$). Overall, $1/2$ of all offspring are expected to be colour-blind.

(c) Males are hemizygous for X-linked genes (they have only one X chromosome, paired with a Y that lacks the corresponding gene), so a single copy of the recessive allele is sufficient to produce the colour-blind phenotype; females have two X chromosomes, so they need two copies of the recessive allele (homozygous recessive) to be affected, as one dominant allele on their other X chromosome is enough to give normal vision, making females less likely to display the recessive phenotype.

Quick check answer

C. With independent assortment of two heterozygous gene pairs, the classic dihybrid phenotype ratio is 9:3:3:1 (9 showing both dominant traits, 3 and 3 showing one dominant/one recessive trait each, and 1 showing both recessive traits).

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.